

Multi-Temporal Mapping of Forest Disturbance and Recovery

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1. Introduction

Forest ecosystems are among the most ecologically significant and carbon-rich biomes on Earth. In India, the Eastern Ghats constitute a discontinuous chain of hill ranges along the eastern coast, harbouring remarkable biodiversity and supporting millions of tribal communities. The Alluri Sitaramaraju (ASR) district of Andhra Pradesh, situated within this mountain chain, is characterised by approximately 64% forest cover, encompassing dry and moist deciduous forests across a rugged and mountainous terrain spanning 7,838 km² (District Survey Report, 2023). This highly forested landscape is under increasing pressure from anthropogenic disturbances and climate variability, making it an urgent subject for scientific monitoring and conservation planning.

Forest disturbances — caused by fire, shifting cultivation (podu), logging, road expansion, and drought — degrade ecosystem function, reduce carbon stocks, and threaten species survival. Understanding the spatial and temporal patterns of these disturbances and the subsequent recovery trajectories of forest vegetation is critical for designing effective conservation and land management strategies. Multi-temporal remote sensing offers a powerful, cost-effective approach to continuously monitor large forest landscapes at high spatial resolution, enabling detection of subtle and significant changes over time.

1a. Literature Survey

A substantial body of scientific literature has examined forest disturbance and recovery using remote sensing techniques across tropical and subtropical ecosystems. Remote sensing-based monitoring has become an important tool for studying vegetation dynamics, wildfire occurrence, land degradation, and ecosystem recovery over large spatial and temporal scales. However, compared to the Western Ghats, the Eastern Ghats region has received relatively less scientific attention, and detailed multi-temporal analyses of wildfire dynamics in the Alluri Sitaramaraju (ASR) district remain limited.

The Normalized Difference Vegetation Index (NDVI), derived from multispectral satellite imagery, is one of the most widely used indicators of vegetation greenness, biomass density, and ecosystem productivity. NDVI time-series analysis has been widely applied to monitor forest degradation, seasonal vegetation cycles, drought stress, and post-disturbance recovery (Tucker, 1979). MODIS datasets, with their frequent revisit times and long-term observations, are commonly used for regional-scale vegetation monitoring.

For wildfire detection, the NASA Fire Information for Resource Management System (FIRMS) provides near real-time active fire data based on MODIS and VIIRS thermal anomaly products. FIRMS datasets have been used to analyse fire seasonality, spatial clustering, and interannual variability across South Asian forests (Roy et al., 2008). In India, the Forest Survey of India (FSI) publishes India State of Forest Reports (ISFR), which include state-level fire calendar information useful for validating seasonal fire patterns.

Burn severity assessment is commonly performed using the Normalized Burn Ratio (NBR) and its differenced form, dNBR. Developed by Key and Benson (2006), dNBR uses Near-Infrared (NIR) and Shortwave Infrared (SWIR-2) reflectance before and after fire events to quantify burn severity and vegetation loss. Landsat imagery, with its 30 m spatial resolution and long-term archive, is widely used for such analyses.

Wildfire risk assessment using the Analytic Hierarchy Process (AHP) has gained importance as a multi-criteria decision-making approach. Bhuyan et al. (2025) applied an AHP-weighted model in the Malkangiri district of Odisha using factors such as Land Surface Temperature (LST), elevation, NDVI, moisture index, land use/land cover (LULC), aspect, wind speed, and slope. Similarly, Sultan et al. (2025) demonstrated wildfire risk modelling using Google Earth Engine (GEE) for reserved forests in the Vellore district, highlighting the effectiveness of cloud-based geospatial analysis.

Several studies have also focused on biodiversity assessment in Eastern Ghats. Aditya and Ganesh (2017) documented biodiversity patterns and threatened species in Eastern Ghats forests, while Ray et al. (2020) analyzed wildlife distribution and habitat utilization across the region. Mahesh et al. (2022) studied amphibian and reptile diversity and discussed habitat associations based on IUCN conservation status. These studies indicate that integrating biodiversity information with remote sensing-derived disturbance indicators can improve understanding of habitat degradation and ecosystem recovery.

Trend analysis methods such as the Mann-Kendall test and Sen's slope estimator are widely used for analyzing environmental time-series data. These non-parametric methods are suitable for satellite-derived datasets and have been applied to analyze trends in fire occurrence, NDVI variability, and climate conditions (Kendall, 1975; Sen, 1968).

Although previous studies have addressed wildfire monitoring, vegetation analysis, biodiversity assessment, and risk modelling independently, a comprehensive multi-temporal assessment integrating wildfire danger indices, burn severity mapping, temporal

trend analysis, and biodiversity habitat assessment for the ASR district has not been extensively explored. This study attempts to address that gap by integrating multi-source satellite datasets, including Landsat 8/9, MODIS, and ERA5-Land data, to analyze forest disturbance and recovery patterns in the Eastern Ghats during 2018–2024.

1b. Objectives of the Study:

The primary objective of this study is to analyze forest disturbance and recovery patterns in the Alluri Sitaram Raju (ASR) district of the Eastern Ghats using multi-temporal satellite remote sensing data from 2018 to 2024.

The specific objectives of the study are:

1. To analyze temporal patterns of forest fire occurrence in the ASR district using MODIS FIRMS active fire datasets.
2. To assess vegetation condition, fuel availability, and moisture stress using spectral indices such as NDVI, NDMI, NBR, and dNBR derived from MODIS and Landsat imagery.
3. To identify wildfire-prone regions by developing a composite wildfire risk model using the Analytic Hierarchy Process (AHP) and multiple environmental, climatic, topographic, and anthropogenic factors.
4. To evaluate burn severity and post-fire vegetation recovery using multi-temporal Landsat imagery.
5. To perform temporal trend analysis of wildfire activity and vegetation recovery using statistical methods such as the Mann-Kendall test and Sen's slope estimator.
6. To identify repeat-burning hotspots and examine long-term disturbance patterns across the study area.
7. To assess the potential impacts of forest disturbance on biodiversity and habitat conditions for selected species in the Eastern Ghats landscape.
8. To support sustainable forest management and conservation planning using remote sensing and geospatial analysis techniques.

1c. Study Area

The study was conducted in the Alluri Sitaram Raju (ASR) district, located in the Eastern Ghats region of Andhra Pradesh, India, approximately between 17°N and 18°N latitude. The district is characterized by mountainous and rugged terrain with extensive forest cover and significant ecological diversity.

The ASR district contains approximately 7,838 km² of forest area, accounting for nearly 64% of the total district area. The dominant vegetation types include dry deciduous and moist deciduous forests, which support a wide range of flora and fauna. The region forms part of an important biodiversity-rich landscape within the Eastern Ghats and includes ecologically sensitive habitats associated with the Papikonda forest region.

The area experiences frequent forest disturbances caused by both natural and anthropogenic factors, including seasonal forest fires, shifting cultivation, logging activities, grazing pressure, road expansion, and climate variability. These disturbances can significantly influence vegetation conditions, ecosystem stability, and wildlife habitats.

Due to its ecological importance, high forest cover, and recurring wildfire activity, the ASR district provides a suitable study area for analyzing forest disturbance, wildfire risk, and post-fire vegetation recovery using multi-temporal remote sensing techniques.

2. Materials and Methods

2.1 Satellite and Ancillary Data

This study utilized multi-source satellite, climatic, topographic, and ancillary geospatial datasets to analyze forest disturbance, wildfire risk, and vegetation recovery patterns in the Alluri Sitaram Raju (ASR) district during 2018–2024. Satellite datasets were processed using Google Earth Engine (GEE) along with standard GIS and Python-based analysis tools. The selected datasets provided information related to vegetation condition, moisture stress, fire occurrence, climatic variability, terrain characteristics, and anthropogenic influence.

Dataset / Source	Parameters Used	Spatial Resolution	Purpose
MODIS MOD13Q1	NDVI	250 m	Vegetation condition and fuel analysis
MODIS FIRMS		1 km	

	Active fire detections		Fire occurrence and temporal fire analysis
Landsat 8/9 Surface Reflectance	NDMI, NBR, dNBR	30 m	Burn severity and vegetation recovery analysis
ERA5-Land	Temperature, humidity, wind speed	~9 km	Weather danger analysis
SRTM DEM	Elevation, slope, aspect	30m	Topographic analysis
Oxford Accessibility Dataset	Travel-time accessibility	100m	Anthropogenic activity analysis
GPW v4 / WorldPop	Population density	100 m – 1 km	Human disturbance assessment
ESA / APSAC LULC Data	Land use/land cover	10–30 m	Wildfire risk modelling
FSI ISFR Reports	Fire season calendars	State level	Fire season validation

2.2 Fire Season Determination (Step 1)

Accurate identification of the active fire season was necessary for selecting appropriate pre-fire and post-fire imagery windows for subsequent analysis. Two independent approaches were used to determine the fire season in the Alluri Sitaram Raju (ASR) district.

The first approach utilized MODIS FIRMS active fire data available through Google Earth Engine (GEE). Monthly and bi-weekly fire pixel counts were extracted for the ASR district for the period 2019–2023. Fire detections were filtered using thermal anomaly

thresholds to identify active fire events, and temporal trends in fire occurrence were analyzed. The resulting time-series consistently showed increased fire activity during the February–April period, with peak fire occurrence observed during March.

The second approach used fire season information from the India State of Forest Reports (ISFR) published by the Forest Survey of India (FSI). The state-level fire calendars for Andhra Pradesh also identified February to April as the peak wildfire period in the Eastern Ghats region.

Both approaches showed strong agreement and confirmed February 15 to April 30 as the primary fire season for the ASR district. Based on this result, pre-fire imagery was selected from October 2022 to January 2023, while post-fire imagery was selected from May to July 2023 for burn severity and vegetation recovery analysis.

2.3 Spectral Indices and Danger Indices (Steps 2–4)

2.3.1 NDVI – Normalized Difference Vegetation Index

The Normalized Difference Vegetation Index (NDVI) was computed from MODIS MOD13Q1 16-day composite imagery at 250 m spatial resolution and resampled to 30 m using bilinear interpolation for integration with Landsat-derived layers.

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

NDVI represents vegetation greenness and biomass density. Higher NDVI values indicate dense and healthy vegetation with greater fuel availability, whereas lower NDVI values indicate sparse vegetation or already-burned regions. Pre-fire NDVI was computed for October 2022–January 2023, while post-fire NDVI was computed for May–July 2023.

2.3.2 NDMI – Normalized Difference Moisture Index

The Normalized Difference Moisture Index (NDMI) was derived from Landsat 9 Surface Reflectance imagery using Near-Infrared (NIR, Band 5) and Shortwave Infrared (SWIR-1, Band 6) bands at 30 m spatial resolution.

$$\text{NDMI} = (\text{NIR} - \text{SWIR1}) / (\text{NIR} + \text{SWIR1})$$

NDMI represents vegetation moisture content and canopy water stress. Lower NDMI values indicate dry and moisture-stressed vegetation with higher fire susceptibility. Cloud and shadow masking were applied using the QA_PIXEL band before image compositing.

2.3.3 FDI – Fuel Danger Index

The Fuel Danger Index (FDI) integrates vegetation density and fuel moisture conditions using NDVI and NDMI layers at 30 m spatial resolution.

$$\text{FDI} = \text{NDVI} + \text{NDMI}$$

The layers were normalised prior to integration to ensure consistent weighting. Areas with high FDI values represent dense and dry vegetation with high fire spread potential.

2.3.4 ADI – Activity Danger Index

The Activity Danger Index (ADI) was used to represent anthropogenic fire pressure within the study area. The index incorporated accessibility and population density datasets as indicators of human activity and disturbance.

$$\text{ADI} = (\text{ADI road} + \text{ADI settlement}) / 2$$

Areas located near roads, settlements, and forest-edge regions recorded higher ADI values, indicating increased likelihood of human-induced fire occurrence.

2.3.5 WDI – Weather Danger Index

The Weather Danger Index (WDI) was derived from ERA5-Land monthly climatic datasets using temperature, relative humidity, and wind speed averaged over the February–April 2023 fire season.

$$\text{WDI} = (\text{Tcls} + \text{RHcls} + \text{WScls}) / 3$$

where:

- Tcls = temperature danger class
- RHcls = relative humidity danger class
- WScls = wind speed danger class

Each variable was normalized and classified before integration. Higher WDI values indicate weather conditions favorable for wildfire ignition and spread.

2.3.6 TDI – Topographic Danger Index

The Topographic Danger Index (TDI) was derived from the USGS SRTM Digital Elevation Model using slope and aspect layers at 30 m spatial resolution.

$$\text{TDI} = (\text{Slope} + \text{Aspect}) / 2$$

The slope and aspect layers were normalized prior to integration and clipped to the ASR district boundary. Higher TDI values indicate terrain conditions favorable for rapid fire spread.

2.3.7 NBR and dNBR – Burn Severity Analysis

The Normalized Burn Ratio (NBR) was computed from Landsat 9 Surface Reflectance imagery using Near-Infrared (NIR, Band 5) and Shortwave Infrared (SWIR-2, Band 7) bands.

$$\text{NBR} = (\text{NIR} - \text{SWIR2}) / (\text{NIR} + \text{SWIR2})$$

Pre-fire NBR was computed using imagery from November–December 2022, while post-fire NBR was computed using imagery from May–June 2023. Burn severity was estimated using differenced NBR (dNBR).

$$\text{dNBR} = \text{NBR}_{\text{pre}} - \text{NBR}_{\text{post}}$$

Higher dNBR values indicate greater burn severity and vegetation damage. Cloud masking using the QA_PIXEL band was applied before differencing. The resulting dNBR layer was classified into Unburned, Low, Moderate, and High severity classes.

2.4 AHP Composite Wildfire Risk Model (Step 5)

A composite wildfire risk map was generated using the Analytic Hierarchy Process (AHP) Weighted Linear Combination approach following the methodology proposed by Bhuyan et al. (2025). The model integrated eight wildfire risk factors at 100 m spatial resolution: Land Surface Temperature (LST), elevation, NDVI, NDMI, land use/land cover (LULC), aspect, wind speed, and slope.

A pairwise comparison matrix was constructed to assign relative importance weights to each factor based on wildfire susceptibility. The weighting scheme followed the approach described by Bhuyan et al. (2025). The consistency of the matrix was evaluated using the Consistency Ratio (CR).

$$\text{CI} = (\lambda_{\text{max}} - n) / (n - 1)$$

$$\text{CR} = \text{CI} / \text{RI}$$

The calculated values ($\lambda_{\text{max}} = 8.7699$, $\text{CI} = 0.1100$, $\text{CR} = 0.078$) satisfied the acceptable consistency condition ($\text{CR} < 0.1$), indicating logical consistency in the pairwise comparison matrix.

The composite wildfire risk index was computed as:

$$\text{Risk} = \sum (w_i \times X_i), i = 1 \text{ to } n$$

where w_i represents the AHP weight and X_i represents the normalized value of each wildfire risk factor.

The resulting continuous wildfire risk surface was classified into five categories: Very Low, Low, Moderate, High, and Very High risk. The final risk map was validated by overlaying MODIS FIRMS active fire detections observed during the February–April 2023 fire season.

2.5 Temporal Analysis (2018–2024)

A systematic temporal analysis was conducted to investigate interannual wildfire dynamics, vegetation variability, and post-fire recovery patterns in the ASR district during 2018–2024. Five analytical components were used to address key scientific questions related to wildfire behavior and ecosystem response.

Q1 – Trend Analysis (Mann-Kendall Test)

The non-parametric Mann-Kendall trend test and Sen's slope estimator were applied to annual FIRMS fire pixel totals and five environmental indices (dNBR, LST, NDVI, WDI, and FDI) for the period 2018–2024 ($n = 7$). The Mann-Kendall statistic (S) and Sen's slope (β) were calculated to identify monotonic trends, while two-tailed p -values were evaluated at a significance level of $\alpha = 0.05$.

Q2 – Anomaly Analysis

Annual index anomalies were normalized relative to the 7-year mean baseline to compare above-average and below-average fire years across multiple environmental indicators. Z-score analysis was performed on annual FIRMS fire pixel counts to rank years based on wildfire severity and anomaly magnitude

Q3 – Driver Correlation Analysis

Pearson correlation coefficients (r) were calculated between annual FIRMS fire pixel counts and nine environmental variables: Land Surface Temperature (LST), ERA5 air temperature, relative humidity, wind speed, NDVI, NDMI, FDI, WDI, and dNBR.

Q4 – Post-Fire Vegetation Recovery

Three seasonal NDVI windows were extracted annually from MODIS MOD13Q1 imagery at 250 m spatial resolution: pre-fire (January), fire season (February–April), and post-monsoon (August–October). Vegetation recovery rate was estimated using:

$$R_y = |\text{NDVI-pre} - \text{NDVI-peak}| / (\text{NDVI-post} - \text{NDVI-peak}) \times 100$$

where NDVI-pre, NDVI-peak, and NDVI-post represent pre-fire, peak-fire, and post-monsoon NDVI values, respectively.

Q5 – Fire Return Interval (FRI) Mapping

Binary fire detection layers were generated annually from MODIS FIRMS active fire data and stacked across all seven years. Pixel-wise summation produced an integer Fire Return Interval (FRI) map ranging from 0–7 at 1 km spatial resolution, representing the number of years in which fire was detected at each location.

All data processing and analysis were performed using the Google Earth Engine (GEE) Python API, Python libraries including NumPy, SciPy, Matplotlib, and PyMannKendall, along with standard GIS techniques.

2.6 Biodiversity and Species Analysis

Species occurrence records were obtained from the Global Biodiversity Information Facility (GBIF) and published biodiversity studies related to the Eastern Ghats region (Aditya & Ganesh, 2017; Ray et al., 2020; Mahesh et al., 2022). Records were filtered to retain only observations located within the ASR district boundary and containing valid georeferenced coordinates. Species were further classified based on IUCN Red List conservation categories (CR, EN, VU, and NT) and taxonomic groups including mammals, birds, reptiles, flora, and other ecological groups.

Habitat condition and vegetation dynamics for each species were assessed using MODIS NDVI time-series data extracted at georeferenced habitat centroid locations for the period 2018–2024 using Google Earth Engine (GEE). Vegetation disturbance during the major fire period (2018–2020) was quantified using NDVI change ($\Delta NDVI$), while post-fire habitat recovery was evaluated using linear regression slopes of NDVI trends during 2020–2024.

The Kruskal–Wallis non-parametric statistical test was applied to compare habitat disturbance patterns across different IUCN conservation categories. In addition, a Conservation Priority Index (CPI) was developed by integrating IUCN conservation status, ecological importance, and occurrence frequency across multiple literature sources. The CPI was used to identify species requiring priority conservation attention within the ASR forest landscape.

3. Results and Discussion

3.1 Fire Season Determination

Analysis of MODIS FIRMS monthly fire pixel counts for the ASR district (2019–2023) consistently showed fire activity concentrated in February–April, with March as the single-month peak. The 5-year average fire detections for March and April reached 977 and 988.6 fire pixels respectively — representing an order-of-magnitude increase relative

to the monsoon months (June–September: ≈ 0 fire pixels). The February–2023 bi-weekly time series confirmed a sharp pulse peaking in mid-March 2023, followed by a secondary peak in early April. FSI fire risk scores for Andhra Pradesh from ISFR 2021 and 2023 independently confirmed February–April as the peak fire-risk window, with ASR district reaching the maximum risk score (5 of 5) in February–March, consistent with the MODIS-derived finding. Both data streams converged on February 15 – April 30 as the fire season, which was adopted as the standard analysis window for all subsequent indices. This is shown in Fig1, Fig2 and Fig3.

The 2020 fire season was a notable exception, recording the lowest fire pixel counts across the study period, coinciding temporally with COVID-19 pandemic mobility restrictions that reduced human access to forests. This observation is consistent with findings from similar studies across South Asian forests during 2020 (Vadrevu et al., 2021) and underscores the importance of anthropogenic activity as a fire ignition driver in the region.

3.2 Vegetation Indices – NDVI, NDMI, and FDI Maps

The pre-fire NDVI map (October 2022–January 2023) revealed high vegetation greenness (NDVI 0.5–0.75) across the forested interior of the ASR district, reflecting the dense deciduous and semi-evergreen canopy in a post-monsoon leaf-on state. Lower NDVI values (≤ 0.3) were concentrated in the southwestern fringe near settlements and previously disturbed agricultural land. These patterns are consistent with the known 64% forest cover of the district.

The NDMI map highlighted localised patches of moisture-stressed vegetation (NDMI < -0.2 , indicated in red) in the northeastern portion of the district (near Visakhapatnam border) and scattered across the western valleys. These areas of low canopy moisture corresponded spatially with fire hotspots detected by FIRMS, corroborating the role of fuel dryness as a fire precondition. The western forested interior displayed uniformly high NDMI (> 0.2), indicating moist deciduous vegetation with lower ignition susceptibility.

The Fuel Danger Index map showed a broadly high FDI across the entire district, with the interior forested blocks recording the highest combined values. This pattern reflects the dual contribution of dense vegetation (high NDVI) and moderate pre-fire moisture stress (moderate-low NDMI) across the landscape. The highest FDI scores were clustered in the central and western valleys, which are also areas of highest fire activity according to FIRMS records.

This is in reference to Fig 4 to Fig 8.

3.3 Activity and Topographic Danger Indices

The Activity Danger Index (ADI) map revealed a heterogeneous pattern of human disturbance pressure across the district. The highest ADI values (Class 4–5, orange-red) were concentrated in the western and central portions of the district, particularly around the Bhadrachalam-adjacent zones and along major road corridors, reflecting high accessibility and population density in these areas. The remote northeastern forest blocks (Class 1–2, dark green) showed very low human activity pressure. This spatial pattern aligns with known tribal land-use patterns where podu cultivation and fire use are concentrated in accessible mid-elevation terrain.

The Topographic Danger Index (TDI) reflected the district's rugged terrain, with mixed slope-aspect combinations producing moderate TDI values (yellow-olive tones) across most of the landscape. The higher-elevation ridges exhibited greater topographic fire risk due to steeper slopes that accelerate fire spread. South-facing slopes in the western valleys recorded the highest TDI values. This pattern indicates that topography moderates but does not dominate the fire risk landscape relative to fuel and meteorological factors.

This is in reference to Fig 4 to Fig 8.

3.4 Burn Severity – Pre-Fire NBR, Post-Fire NBR, and dNBR

The pre-fire NBR map (November–December 2022) showed predominantly high NBR values (0.4–0.6, dark green) across the district's forested interior, consistent with dense, healthy vegetation in post-monsoon condition. The post-fire NBR (May–June 2023) displayed a marked shift: large areas in the western and central portions of the district showed reduced NBR values, while the northeastern sector retained moderate NBR, indicating less fire impact.

The dNBR map revealed widespread low-to-moderate burn severity across the western and southern portions of the district (dNBR 0.1–0.4), with patches of high severity (dNBR > 0.4) in the southwestern valleys near the Bhadrachalam area. The northeastern sector showed near-zero or slightly negative dNBR, consistent with vegetation greenup from early monsoon onset post-fire. The spatial pattern of dNBR hotspots corresponded closely with the high-ADI zones identified in Section 3.3, confirming that human-accessible areas with high activity pressure are most intensely burned.

Compared to Bhuyan et al. (2025) who found that Malkangiri district (a geographically adjacent study area) had predominantly low-to-moderate burn severity with limited high-severity patches, the ASR district showed comparable severity patterns, reflecting similar ecological and climatic conditions across the Eastern Ghats landscape.

This is in reference to Fig 4 to Fig 8.

3.5 AHP Composite Wildfire Risk

The AHP composite wildfire risk map classified the ASR district into five risk zones. The risk distribution was approximately equal across zones, with each class accounting for roughly 20% of the district area, indicating a spatially distributed fire risk landscape with no single dominant risk class.

Comparison with Bhuyan et al. (2025), who found that Malkangiri (the reference study area) had Very Low (40.91%) and Low (33.06%) risk classes dominating, with zero area classified as Very High risk, reveals a striking difference: ASR district has a substantially higher proportion of High and Very High risk zones. This contrast is attributable to ASR's higher population density and greater agricultural encroachment into forest margins, lower mean elevation relative to Malkangiri's more remote terrain, and the more frequent fire activity documented in FIRMS records. The LULC analysis confirms that Tree Cover (8,766.2 km², Class 5 – highest fire risk) dominates the district area, directly elevating the overall composite risk relative to the Malkangiri landscape.

Validation with MODIS FIRMS fire detections for February–April 2023 confirmed strong spatial correspondence between the High and Very High AHP risk zones and observed fire pixel locations, particularly in the western and central portions of the district. Fire points predominantly fell within Class 4–5 zones, validating the model's predictive capacity. (Fig 9 to 23)

3.6 Temporal Fire Analysis (2018–2024)

3.6.1 Annual Fire Activity (Q1 – Trend Analysis)

Annual FIRMS fire pixel totals for the ASR district from 2018 to 2024 ranged from 3,227 (2020) to 7,942 (2021), with a 7-year mean of approximately 5,921 fire pixels per year. The Mann-Kendall trend test returned a non-significant result (Sen's slope = -76.84 pixels/year, $p = 1.000$), indicating no statistically significant monotonic trend in fire activity over the study period. This finding is consistent with results from similar 7-year remote sensing time series in Indian forest landscapes, where year-to-year variability driven by rainfall anomalies typically masks longer-term trends. (Fig 24)

3.6.2 Anomaly Analysis (Q2 – Severe Years)

Normalised anomaly analysis confirmed 2021 as the most severe fire year ($z = +1.11$, rank 1 of 7) and 2019 as the second most severe ($z = +1.04$, rank 2). Both years showed above-mean fire pixels, above-mean LST, and below-mean NDMI (drier conditions). The 2020 fire season was the least active across the study period ($z = -1.47$, rank 7), with the lowest fire pixel count of any year. This dramatic suppression is attributed to COVID-19-related mobility restrictions (national lockdown: March–June 2020) which severely curtailed human access to forests and reduced anthropogenic ignition sources. The 2020 anomaly, with simultaneously low fire activity and slightly elevated NDVI (vegetation

greener than average), provides a natural experiment confirming that human activity is the dominant ignition source in this landscape.(Fig 25)

The anomaly heatmap across all six indices reveals an interesting pattern: high-fire years (2018, 2019, 2021) tend to show above-average LST and below-average NDMI, whereas low-fire years (2020, 2022, 2023) show the inverse pattern. This consistent relationship across multiple independent datasets strengthens confidence in the direction of association between atmospheric dryness and fire intensity.

3.6.3 Driver Correlation Analysis (Q3 – What Drives Bad Fire Years?)

Pearson correlation analysis identified NDMI (vegetation moisture) as the strongest predictor of interannual fire intensity ($r = -0.767$, $p < 0.05$), followed by FDI ($r = -0.666$), relative humidity ($r = -0.553$), and NDVI ($r = -0.469$). In contrast, ERA5 air temperature ($r = +0.564$) and WDI ($r = +0.564$) showed positive, though statistically non-significant, associations with fire activity. Slope and wind speed showed negligible correlation ($|r| < 0.20$). (Fig 26)

The dominant role of NDMI confirms that pre-fire canopy moisture is the primary modulator of fire intensity in the ASR district. When the vegetation is drier (lower NDMI) heading into the fire season, fires burn more extensively. This finding aligns with global literature on fire-moisture relationships in tropical deciduous forests (Nepstad et al., 2004; Cochrane, 2003). It also has direct implications for early warning: NDMI monitoring from Landsat imagery during November–January (post-monsoon drying period) can provide a 2–3 month lead-time forecast of upcoming fire season severity.

The positive correlation with air temperature (higher temperature → more fire) and the negative correlation with relative humidity (lower humidity → more fire) are physically consistent with fire weather theory and support the validity of the WDI formulation used in the AHP model. The fact that NDMI outperforms pure meteorological drivers suggests that vegetation-integrated moisture — incorporating both atmospheric and soil water supply signals — is a more sensitive fire predictor than surface air measurements alone in this landscape.

3.6.4 Post-Fire Vegetation Recovery (Q4)

All seven years studied (2018–2024) showed post-monsoon NDVI recovery rates exceeding 100%, indicating full and often supranormal vegetation rebound by August–October following each fire season. The highest recovery rate was observed in 2018 ($R = 227.8\%$), and the lowest in 2019 ($R = 144.2\%$) — notably, the second-highest fire activity year. Recovery rates in 2023 were also high ($R = 210.8\%$), suggesting rapid ecosystem resilience following the confirmed fire season.(Fig 27)

The consistently above-100% recovery rates across all years are explained by the strong monsoon-driven vegetation rebound typical of deciduous forest ecosystems in peninsular

India: the post-monsoon NDVI regularly exceeds the pre-fire January NDVI because the monsoon delivers substantial moisture that drives both early-season greenup and post-fire regrowth simultaneously. This pattern is consistent with recovery dynamics documented in dry deciduous forests of central India (Mondal & Southworth, 2010).

The weakest recovery in 2019 (despite the second-highest fire pixel count) suggests that particularly intense or repeated burning in that year may have suppressed biomass recovery relative to what the monsoon would otherwise enable. This interpretation is supported by the relatively low post-fire NDVI values observed at severely burned sites, indicating that high-severity fires create longer-lasting canopy gaps than low-severity surface fires.

3.6.5 Fire Return Interval (Q5 – Repeat Burning)

Fire return interval analysis revealed that 8,631 km² (approximately 68.3% of the total analysed area) burned at least once during 2018–2024. More critically, 7,748.6 km² — equivalent to 63.2% of the ASR district — burned at least twice during the 7-year period. Areas burning three or more times covered 1,946.3 km², with 48.2 km² burning in all 7 years studied. The FRI map shows that repeat-burning hotspots are concentrated in the western and central valleys, aligned with high ADI (human activity) and moderate-high FDI zones. (Fig 28)

The prevalence of repeat burning across more than 60% of the district is ecologically significant. Repeated high-frequency fires inhibit forest regeneration by killing seedlings and saplings before they can reach fire-resistant bark thickness (Cochrane, 2003). They also degrade soil structure, reduce litter accumulation, and progressively push ecosystems toward fire-adapted grassland or scrub states — positive feedback known as the ‘grass-fire cycle’ (D’Antonio & Vitousek, 1992). The high FRI documented here constitutes a major obstacle to long-term forest recovery in the ASR district and highlights the urgent need for targeted firebreak establishment, controlled-burn management, and community-based fire prevention strategies in the most frequently burned western zones.

3.7 Biodiversity and Species Habitat Analysis

3.7.1 Species Distribution and Habitat Condition

A total of 31 threatened and near-threatened species were mapped across the ASR district, encompassing mammals (e.g., Asian Elephant, Bengal Tiger, Sloth Bear, Indian Pangolin), birds (Great Hornbill, Alexandrine Parakeet, B&R Flycatcher), reptiles (King Cobra, Indian Rock Python, Mugger Crocodile, Gharial), flora (Indian Kino Tree, Bedome’s Cycad, Indian Frankincense), amphibians (Purple Frog, Amboli Toad), and aquatic species (Mahseer, Goonch Catfish). Species observations are clustered in the Papikonda National Park region and along major river corridors (Godavari and its

tributaries), consistent with the dependency of most species on intact, dense forest and riparian ecosystems.

Habitat NDVI time series (2018–2024) extracted at species occurrence coordinates revealed distinct patterns across species. The Asian Elephant showed the most stable habitat NDVI (consistently above 0.75), indicating that elephant habitat in the Papikonda NP core zone remained largely undisturbed by fire. Indian Pangolin habitat showed moderate NDVI fluctuations (0.58–0.70), suggesting periodic fire-related disturbance but good recovery. Mahseer (an aquatic species dependent on riparian forest shade) showed the largest NDVI variability and the weakest recovery slope of all species (NDVI recovery slope = $-0.0097/\text{year}$), indicating progressive riparian habitat degradation that likely threatens water temperature and turbidity conditions critical for the species' reproduction.

3.7.2 Habitat Disturbance vs. Recovery Analysis

The disturbance exposure vs. recovery scatter plot revealed that most species habitats fell in the low-disturbance, poor-to-moderate recovery quadrant, with Mahseer showing the extreme case of high NDVI loss ($d\text{NDVI } 2018\text{--}2020 = -0.013$) combined with the weakest post-fire recovery (slope = -0.0097). Indian Rosewood (a timber tree facing unsustainable harvesting) showed similarly low recovery. In contrast, Indian Rock Python and Smooth-coated Otter habitats showed low disturbance loss combined with moderate recovery, placing them in a more favourable ecological position. (Fig 29 and Fig 30)

The Kruskal-Wallis test comparing habitat disturbance across IUCN status categories returned $H = 3.267$, $p = 0.071$ (Fig 31). While this result is statistically non-significant at the conventional $\alpha = 0.05$ threshold, the p-value approaching significance suggests a trend towards greater habitat loss in Endangered species habitats compared to Vulnerable and Near Threatened categories. The non-significance likely reflects the small sample sizes per category and substantial within-category spatial heterogeneity in habitat conditions. A larger dataset with more species records per category would be required to achieve statistical power sufficient to detect such differences.

3.7.3 Conservation Priority Index (CPI) Ranking

The Conservation Priority Index ranking, integrating IUCN status, ecological importance, and multi-literature occurrence, identified the following species as highest priority for conservation intervention in the ASR district ($\text{CPI} \geq 5.5$): Red-crowned Roofed Turtle (CR), Gharial (CR), Dhole (EN), Bengal Tiger (EN), Wild Jamun/Malabar Sarai (Eastern Ghats endemics, EN), Purple Frog (EN), Lantern Flower (CR), and Bedome's Cycad (CR). These species represent the most extinction vulnerable taxa with the greatest dependence on intact forested landscapes in the district.

Per-species habitat recovery slope analysis identified Mahseer, Indian Rosewood, Alexandrine Parakeet, and Asian Elephant as having the most negative NDVI recovery trends (weakest habitat improvement 2020–2024), despite these species occupying habitats with varying fire exposure. This underscores that habitat degradation in the ASR district is driven not only by fire but also by other stressors including selective timber harvesting, riparian modification, and agricultural encroachment (Fig 32 and Fig 33).

3.8 Validation of AHP Wildfire Risk Model Using MODIS FIRMS Active Fire Data

3.8.1 Validation Approach

The composite AHP wildfire risk map was validated by comparing the spatial distribution of predicted risk classes against independently observed MODIS FIRMS active fire detections. FIRMS provides thermal anomaly-based fire detections derived from the MODIS Terra and Aqua sensors at 1 km spatial resolution, processed through NASA's Land, Atmosphere Near real-time Capability for EOS (LANCER) system. These detections are entirely independent of the AHP model inputs, making them a robust external validation dataset.

Active fire pixels were extracted for the peak fire season (February–April) of each study year (2019–2023) within the ASR district boundary. The validation rests on the premise that if the AHP risk model is spatially accurate, a statistically significant proportion of observed fire detections should fall within the High (Class 4) and Very High (Class 5) risk zones.

3.8.2 Annual FIRMS Fire Detection Counts

From Figures 34–38 Annual FIRMS fire pixel totals for the ASR district over the 2019–2023 period showed strong interannual variability. The year 2021 recorded the highest fire activity, while 2020 recorded a pronounced suppression consistent with COVID-19 pandemic mobility restrictions that substantially reduced anthropogenic ignition pressure across Indian forests. The year 2019 ranked as the second most severe fire year. These patterns confirm that human accessibility and activity are primary drivers of fire ignition in the ASR landscape.

For validation purposes, the February–April fire pixels for each year were overlaid on the AHP risk map generated for the corresponding year. This year-specific validation accounts for the annual variation in dynamic factors — particularly Land Surface Temperature, NDVI, and vegetation moisture — that shift the spatial risk surface from year to year.

3.8.3 Spatial Concordance Between Risk Zones and Fire Detections

Overlay analysis consistently demonstrated that most FIRMS fire detections fell within the High and Very High AHP risk zones (Classes 4 and 5). The western and central

sectors of the ASR district recorded the highest Activity Danger Index values due to proximity to roads, settlements, and tribal land-use areas — showed the strongest spatial clustering of both fire detections and high-risk model outputs. The northeastern forest blocks, predicted as Low and Very Low risk, recorded negligible fire activity in all study years, further confirming the model's discriminative capacity.

In the high-fire year 2021, fire detections were spatially extensive and well-distributed across the High and Very High-risk zones. In 2020, the low fire count was concentrated in a narrow band of Very High-risk pixels in the western corridor, suggesting that even in a suppressed year, the limited fires that did occur aligned with the highest-risk locations predicted by the model.

3.8.4 Comparison with Reference Study (Bhuyan et al., 2025)

The AHP model developed for ASR district follows the methodology of Bhuyan et al. (2025), who applied the same eight-factor AHP framework to the Malkangiri district of Odisha, a geographically adjacent landscape within the Eastern Ghats. In Malkangiri, Very Low and Low risk classes dominated the spatial distribution (approximately 74% of district area), with no area classified as Very High risk. By contrast, the ASR district exhibited a more balanced and elevated risk distribution, with High and Very High classes together accounting for a substantially larger proportion of the landscape.

This contrast is attributable to three key structural differences: (i) the ASR district has a higher population density and greater agricultural encroachment into forest margins, increasing the Activity Danger Index; (ii) the dominant land cover in ASR is Tree Cover (Class 5 — highest LULC fire risk), whereas Malkangiri has a more heterogeneous land cover mix; and (iii) the ASR district's lower mean elevation reduces the topographic buffering that characterizes Malkangiri's more remote terrain. The validation using FIRMS detections supports the ASR model's elevated risk classification as ecologically and empirically justified rather than a modelling artefact.

3.8.5 Quantitative Validation Summary

The proportion of FIRMS fire pixels falling within High and Very High AHP risk zones was calculated for each study year. Across all five years (2019–2023), the majority of fire detections consistently fell within Classes 4 and 5, with the concordance rate highest during severe fire years (2021, 2019) and slightly reduced during suppressed years (2020, 2022). This pattern is expected: in low-fire years, even the highest-risk areas may record few detections simply because fewer ignitions occur overall, not because the risk model is misclassified.

The Consistency Ratio of the pairwise comparison matrix ($CR = 0.078$) confirmed acceptable consistency ($CR < 0.10$) in the expert weighting scheme, ensuring the AHP weights are logically coherent. Combined with the spatial concordance with FIRMS observations, these results indicate that the AHP composite model provides a reliable spatial characterization of wildfire risk in the ASR district across the 2019–2023 period.

3.8.6 Limitations of the Validation

Several limitations should be acknowledged. FIRMS detections have a spatial resolution of approximately 1 km, which may cause minor spatial misalignment when compared against the 100 m AHP risk surface. Additionally, FIRMS only detects active fire at the moment of satellite overpass; cool or low-intensity fires, and fires burning at night or under cloud cover, may be missed. Finally, FIRMS detections include both escaped agricultural burns and forest fires, and it is not possible to fully separate these categories from thermal anomaly data alone. Despite these limitations, FIRMS remains the most comprehensive, temporally consistent, and spatially explicit active fire dataset available for the study period and region, and its use for AHP validation is standard practice in the literature (Sultan et al., 2025; Bhuyan et al., 2025).

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